# Food Processing and Manufacturing Facility-Class Design Basis

The First Facility-Class Instrument Written for the Food and Agriculture Sector Design Basis

DBA-FA-FP-FC2 — Facility-Class Design Basis, DBA-FA Series

Issued Under: Food and Agriculture Sector Framework (DBA-FA_Sector_Framework v0.4); DBA-MA Parent Framework

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# 1. Purpose and Scope

## 1.1 Purpose

This document sets out the design basis for food processing and manufacturing, the first facility class in the DBA-FA series. The *Food and Agriculture Sector Framework* (DBA-FA_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility: the plant where raw agricultural product becomes food. It names the events a processing plant must withstand, traces how each event reaches the people who eat what the plant makes and the people who depend on it being made, sets the parameters that size the plant's protection, and states what the plant must be able to do when it must stop, when it must hold what it has made, and when it must start again. It describes no particular plant. A design basis for a specific plant, company or commodity starts from this document and adds only what is particular to it.

The framework calls Food and Agriculture a dual-consequence, distributed-web sector: it carries a safety consequence and a supply consequence that are governed differently, and its production is dispersed across many farms but concentrated at processing, where a small number of large plants handle a national share of throughput for their commodity (framework §1.5, §3.3). It names processing as the sector's concentration chokepoint and the point at which a deliberate contamination enters the food supply at scale (framework §3.3), and it proposes two consequence measures, a supply-continuity duration and a contamination consequence measured by exposed population, which it asks not to be collapsed into one (framework §3.6). This document takes both up. On the author's decisions of 14 October 2026 the class is written for all processing commodities at class level, with three reference facilities and with beverage and packaged-food manufacture stated at the depth of the sub-index (Section 2.2); the ammonia refrigeration system of a processing plant is governed here, its release consequence read through the chemical series' shared release model (Section 5.3); and the class proposes three reference events of its own (Section 9.2).

## 1.2 Scope

The class covers the facilities in which raw agricultural product is turned into food, as the framework's register names them: meat and poultry processing, dairy processing, grain milling, beverage manufacturing and packaged-food production (framework §3.3, §8.2). For a plant it covers the receipt and acceptance of raw material, from the live animals at a slaughter plant's pens to the raw milk at a dairy's intake bay; the process itself, with the kill step and the controls that make food safe; the plant's sanitation and its environment, where pathogens persist; the release of finished product, and the hold and recall of product that should not be released; the plant's utilities, its electric supply, water, steam, compressed air and refrigeration; and the control and business systems on which production, traceability and release now depend. It includes the plant's workforce, because a processing plant is among the most labor-intensive facilities in the corpus and stops when its people cannot work.

Three neighbors are separate classes of this series. The farms, feedlots, dairies and the animal and plant health on which supply begins are *Agricultural Production* (DBA-FA-AG-FC1). The cold storage, warehouses, distribution centers and retail that carry food from the plant to the public are *Storage, Distribution and Retail* (DBA-FA-DR-FC3). The inspection systems, the food-safety regulatory framework and disease surveillance are the coordination class, DBA-FA-OpsCo. They are covered here only where the plant depends on them. Where an inspector of the Department of Agriculture is present on the slaughter floor, the inspector's authority is the coordination class's and the plant's response to it is this class's; where finished product leaves the plant's dock, it becomes the distribution class's.

The framework draws boundaries with other series that this class takes as given (framework §6.2). Fertilizer, pesticide and the sanitation chemicals a plant buys are produced in the chemical series; veterinary pharmaceuticals in the healthcare series; and the cold chain beyond the plant is shared with the transportation series. One boundary is drawn here, on the author's decision of 14 October 2026: the anhydrous-ammonia refrigeration system that most large meat, poultry and dairy plants run is the plant's own equipment and is governed by this class, and the consequence of a release from it is read through the shared chemical-release model the chemical series holds in DBA-CHEM-PR-FC1 (*Chemical Production and Synthesis*), at its §4, cited rather than restated (Section 5.3). Section 8.3 states the boundaries.

What makes a facility a member of this class is the same at every site: it takes in what farms produce, makes it into food that will be eaten by people who cannot inspect it, and does so at a scale that makes it, for its commodity, one of few. The particular commodity, process and population differ from plant to plant, and that detail belongs in each plant's own design basis.

## 1.3 The Reference Facilities

Three kinds of plant are used as reference cases. Each was chosen because it shows a different way the output of a processing plant can be lost or made unsafe.

The large beef or pork slaughter and fabrication plant is the governing configuration for the supply axis. A handful of such plants process most of the nation's cattle and hogs, each handling several thousand animals a day, and the live animals upstream cannot wait long for them: cattle and hogs ready for slaughter keep arriving whether or not the plant is open. When the Tyson Fresh Meats plant at Holcomb, Kansas, burned on 9 August 2019, it took out about six thousand head a day, about six percent of United States fed-cattle slaughter capacity, and wholesale boxed-beef prices rose by more than ten percent in the two weeks that followed, as reported (American Farm Bureau Federation, Market Intel, 2019). The plant also runs a large ammonia refrigeration system and employs a workforce in the thousands. The ransomware attack on JBS in May and June 2021, conferred by the framework as DBA-FA-RE-1 (framework §9.1; Section 5.5), stopped plants of this kind across a company whose share of United States beef capacity is about 23 percent on another processor's investor materials, as Bloomberg reported on 2 June

2021; no public body's figure has been found. National cattle slaughter fell 22 percent that day from a week earlier, on the Department of Agriculture's estimates as reported.

The sole-source specialty-nutrition plant makes a product for which there is no ready substitute and few other makers: infant formula, and in particular the specialty and metabolic formulas on which some infants and patients depend entirely. It is the governing configuration for the joint case in which a safety finding becomes a supply failure, because the plant cannot be both stopped for safety and relied on for supply, and the population it serves cannot wait. The shutdown of Abbott Nutrition's plant at Sturgis, Michigan, in February 2022, conferred by the framework as DBA-FA-RE-2 (framework §9.1; Section 5.4), is the case.

The dairy plant with its raw-milk intake receives raw milk daily from many farms, often through collection stations and haulers, pasteurizes and processes it, and ships finished product within days. It is the governing configuration for the safety axis at intake, because what enters at the bay from many sources is pooled before it is processed, and an adulterant introduced upstream of the pool reaches everything made from it. It is also the class's clearest example of a locally sovereign protective function, the flow-diversion device of a high-temperature, short-time pasteurizer (Section 7.2). The adulteration of milk with melamine in China in 2008, conferred by the framework as DBA-FA-RE-3 (framework §9.1; Section 5.6), is the case.

## 1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it governs is the loss or contamination of food at a place where food is made, and that harm runs through the plant, its process, its environment and the people in it, even where the cause is the loss of an information system. In the framework's structure it sits under Food Processing and Manufacturing (FP), the second of the sector's three function classes, and it is the first written (framework §3.1, §3.3). It returns to the physical structure after the two logical classes written before it in the plan for closing the seven sectors, and it follows the structure of *Care Delivery* (DBA-HPH-CD-FC1) and *Chemical Production and Synthesis* (DBA-CHEM-PR-FC1).

## 1.5 Relationship to Governing Documents

This instrument is issued under the Food and Agriculture Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, consequence measures or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form framework §N resolve against the Food and Agriculture Sector Framework in force. The framework inherits the water sector's contamination-consequence structure on the safety axis (framework §1.5, §8.1), and this class reads its safety chain through that structure as the framework directs.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term envelope. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

# 2. The Food Processing and Manufacturing Facility Class

## 2.1 Defining Characteristics

Five characteristics define the class, and each is a reason a processing plant's failure reaches farther than the plant.

The first is concentration. For many commodities a few plants process most of the national supply, and they do so continuously, so that the loss of one is a loss of a measurable share of the nation's supply the same day. The framework calls processing the sector's chokepoint for this reason (framework §3.3).

The second is pooling. A plant combines what it receives from many sources into lots, and what is in one source is then in the lot. A pathogen in one carcass, an adulterant in one tanker of milk, or a contaminated ingredient from one supplier reaches everything made from the pool. Pooling is how a local contamination becomes a national one.

The third is release without inspection by the consumer. The people who eat what the plant makes cannot test it, and the plant's own testing samples rather than examines every unit. The safety of what is released rests on the process controls that make it safe, and on the plant's decision to release it.

The fourth is the perishable inflow. A plant cannot simply stop receiving. Live animals ready for slaughter, raw milk produced twice a day and harvested crops keep coming, and a plant that stops leaves its suppliers with product that loses value by the day and, in the case of milk, by the hour. A stop at the plant is felt upstream as quickly as downstream.

The fifth is labor intensity. Large meat and poultry plants employ thousands of people working close together at a line speed set by the process. Their absence stops the plant as surely as the loss of power, and the spring of 2020 showed it.

## 2.2 The Commodity and Perishability Sub-Index

The framework sub-indexes the sector's classes by commodity type and perishability, so that the design-basis load a facility carries reflects what it handles and how time-critical its cold chain is (framework §3.1, §8.2). This instrument states three bands for the class.

The first band is the sole-source or dominant plant for a commodity on which a vulnerable population depends without substitute: infant formula and medical foods above all, and any product for which the plant is, for some population, the only practical source. Its loss is a safety consequence for that population even when nothing it made was unsafe, because the population cannot be fed otherwise. It carries the full design-basis load on both axes.

The second band is the high-throughput plant handling a perishable commodity that most of the population eats: the large slaughter and fabrication plants for beef, pork and poultry, and the large dairy plants. Its loss is felt in national supply within days and in its suppliers' losses within hours, and a contamination it releases reaches a national population. It carries the full design-basis load on the supply axis and on the safety axis.

The third band is every other processing plant. It is in the class, and the class's provisions apply to it in proportion to the share of supply it carries and the population its products reach.

Beverage manufacturing and packaged-food production are stated at the depth of this sub-index only, on the author's decision of 14 October 2026. Each carries the bands above, a plant making a nutritionally complete product for a dependent population falling in the first, and their design bases are held for a later revision (Section 10.3).

## 2.3 Deployment Typologies and Size Range

The class spans several kinds of plant, and the design basis turns on which a plant is. A slaughter plant receives live animals and takes them through stunning, slaughter, evisceration and chilling under continuous federal inspection, and a fabrication floor breaks carcasses into cuts and grinds; a further-processing plant makes cooked, cured and ready-to-eat products from them. A dairy plant receives raw milk, pasteurizes it and makes fluid milk, cheese, butter or powders; a powder plant may feed formula manufacture. A spray-drying plant makes powdered nutritional products, in which a pathogen that survives in the dry environment of the plant can reach a product that is not cooked again before it is eaten. A mill grinds grain into flour, and a packaged-food plant combines ingredients from many suppliers.

The range is wide. At one end is a slaughter plant handling six thousand cattle a day with a workforce of several thousand; at the other a small plant serving a region. The class is written to fit both, because the consequence of either can reach the population that eats what it makes.

## 2.4 Consequence Context

The class carries both of the framework's consequences, and they are stated separately, as the framework requires (framework §3.6). On the supply axis the consequence is the loss of a share of a commodity's processing capacity, felt as shortage, price and the loss to suppliers of product they cannot sell, and measured by how long the served population can be sustained from what is stored and in transit. On the safety axis the consequence is food that makes people ill, felt as illness and death, measured by the population exposed, and marked, as the water sector found of contamination, by being geographically broad, delayed from the point of introduction, and hard to attribute quickly (framework §1.5).

The two meet in this class more often than in any other. A safety finding at a plant stops the plant, and the stop is a supply event; a supply crisis presses a plant to release what it has, and the pressure is a safety risk. On 17 February 2022 a voluntary recall of powdered infant formula made at Abbott Nutrition's Sturgis plant began, following reports of infants who had consumed it falling ill; the Commissioner later counted four *Cronobacter* cases, in two of which the infection may have contributed to death, and recorded that the plant's causal role was not established; the plant was the only Abbott site making certain specialty formulas, and by the week ending 15 May 2022 the national in-stock rate for infant formula had fallen to nearly 80 percent from 89 to 90 percent before the recall (Food and Drug Administration, testimony of the Commissioner to the Senate Committee on Health, Education, Labor and Pensions, 26 May 2022). In 2008 milk adulterated with melamine reached infant formula and other products of many Chinese dairy companies, and by the end of November 2008, on the Ministry of Health's figures as the Chinese Center for Disease Control and Prevention's papers of 2009 give them, about 294,000 children had been diagnosed with urinary-tract stones, more than 50,000 had been hospitalized, and six deaths had been confirmed. The outward consequence the framework describes, a supply disruption producing shortage and a contamination producing harm at population scale (framework §6.3), is what both chains set off.

# 3. The Facility-Class Design Envelope

## 3.1 Why Processing Is the First Class

Three reasons put this class first. The framework names processing as the sector's chokepoint, the point at which a distributed production system becomes a concentrated, high-consequence facility, and as the entry point of deliberate contamination at scale (framework §3.3), so the sector's design-basis load is greatest here. The plan for closing the seven sectors names it as the sector's first class for that reason. And the record is good: the class's events have been investigated by the Food and Drug Administration and the Department of Agriculture, by Congress, by the Centers for Disease Control and Prevention and by the World Health Organization, so the parameters can be stated against what is known. Agricultural production,

whose governing event is animal and plant disease, is left to its own class, for which the event the plan named here is held (Section 9.2).

## 3.2 The Two Consequence Chains

Every event in the class reaches the population by one of two chains, and the framework's insistence that they not be collapsed is kept here.

Chain U, the supply chain, begins with something that stops a plant from making food: fire or flood, the loss of power, water or refrigeration, an attack on its control or business systems, the loss of its workforce, or a regulatory stop after a safety finding. Output falls; the commodity's national supply falls by the plant's share; suppliers upstream are left with animals, milk or crops they cannot sell; and the served population draws down what is stored and in transit. Its consequence is measured by the supply-continuity duration of framework §3.6 and grows with the time the plant is down and the share it carries. The JBS attack of 2021 is the case.

Chain S, the safety chain, begins with something that makes what the plant releases unsafe: a pathogen in the plant's environment, a failure of the kill step, an adulterant in what the plant received, or a contaminant introduced deliberately. The product is released, distributed and eaten before the hazard is known, and the consequence appears days or weeks later, in people spread across the plant's distribution area. Its consequence is measured by the exposed population and is fixed largely at the moment of release. The melamine adulteration of 2008 is the case on its deliberate side, and Sturgis on its accidental side, as a hazard found in the plant whose causal role in the illnesses is unresolved.

The two chains are joined by the plant's own response. A plant that finds a hazard stops and holds, and Chain S becomes Chain U; a plant whose output is urgently needed is pressed to keep running, and Chain U risks Chain S. The Sturgis case is both, and the class's provisions on holding, recalling and restarting (Section 4) are written for the point where the two meet.

## 3.3 Design Parameters at Class Resolution

Five parameters govern the envelope. Together they are this class's calibration of the two consequence measures the framework proposes (framework §3.6), and Section 10.2 records them for the framework to take up.

The first is the supply share: the fraction of a commodity's national processing capacity the plant carries, and, for a first-band plant, the fraction of the dependent population it is the practical source for. It sets how much of Chain U a single plant's loss produces, and it is the parameter that places a plant in its band. For a cyber event the supply share is counted across the shared system: the combined share of every plant that shares a system whose loss would stop them all, because in 2021 an attack on servers serving one company's systems stopped all of its United States beef plants within a day (*The Ransomware Attack on JBS* F-1, R-1).

The second is the supply-continuity duration: for the population the plant serves, the time it can be sustained from product stored and in transit, and from the spare capacity of other plants, before shortage. The framework proposes the measure (framework §3.6). For a high-throughput perishable commodity it is short, because the modern food system holds little stock; for a first-band product it can be shorter still, because the dependent population has no substitute. The plant's design basis states it, and states the time to restore the plant against it.

The third is the exposed population: the number of people who will eat what a plant releases in the interval between a hazard's introduction and its detection, set by the plant's throughput, the size of its lots, its distribution area and the time to detect. It is the framework's safety measure (framework §3.6), and it is fixed largely at the moment of release.

The fourth is the time to detect and hold: the time from the introduction of a hazard to the plant's hold on everything that may contain it, through its own testing, through a regulator's finding, or through illness reported in the population. It bounds the exposed population. In the Sturgis case the first illness complaint reached the Food and Drug Administration in September 2021 and the recall began on 17 February 2022, about five months later, a delay the Commissioner's testimony attributes in part to the agency's own systems (testimony of 26 May 2022, Appendix A). The time to detect is counted from the first report from inside the plant as well as from the first illness, because in the Sturgis case a whistleblower complaint received in February 2021 preceded the first reported infection (*The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* F-1). It is bounded also by whoever watches the dependent population's health: in 2008 pediatricians saw the kidney stones months before the link to formula was made (*The Adulteration of Milk With Melamine in China* F-2).

The fifth is the time to restart: the time from a stop to the resumption of safe production at the plant's design rate, including the investigation and correction a regulator requires before release can resume. It bounds Chain U after a safety stop. In the Sturgis case a consent decree was entered on 16 May 2022, production restarted in early June, and a severe storm and flooding stopped it again within days, as reported, so that a plant stopped for safety in February was not reliably producing until the summer.

This instrument states the parameters and their governing logic. Their quantitative values for a plant await validation against what plants and regulators already hold: production and shipment records, hazard analyses and food-safety plans, testing records, recall histories and the regulator's inspection findings (Section 10.1). They are not asserted here as numbers, and the figures quoted are the events' own.

# 4. Release, Hold and Restart

## 4.1 Status and Use

Every event in this class reaches the population through what the plant releases or fails to release. The plant's decisions to accept what it receives, to release what it has made, to hold and

recall what it should not have released, and to start again after a stop are the points at which the class governs both chains, and this section states what the class requires of them. It is written as the hospital class writes its receiving function (*Care Delivery* §4), because it is the plant's interface with the population it feeds.

## 4.2 Acceptance: What Enters the Pool

A plant's exposure begins with what it accepts. In 2008, on the account of the Chinese authorities as reported by the World Health Organization, melamine was added to raw milk upstream of the processors, at collection stations and by dealers, to raise its apparent protein content against tests that measured nitrogen, and the processors accepted the milk on those tests. The test measured a proxy for what it was meant to show, and an adulterant chosen to defeat the proxy passed it. The class requires that a first- or second-band plant state, for each raw material it pools, what its acceptance tests show and what they do not, and that it hold a test or a supplier control for each adulterant or contaminant that would defeat them and whose consequence reaches the threshold. It requires that the plant know where each lot of raw material came from, so that an acceptance failure found later can be traced to every lot it entered. Where an acceptance test measures a proxy for the property it is meant to show, as the nitrogen test measured protein, the plant adds a test of the property itself, or one that tells the proxy's legitimate sources from substances that could raise it; testing every batch for the last adulterant found leaves the proxy open to the next (*The Adulteration of Milk With Melamine in China* F-1, R-1). And the stretch between farm and plant belongs to a design basis: collection, hauling and aggregation are placed inside the plant's supplier controls, because in 2008 the adulteration happened at collection stations for which, on the authorities' account, no single agency held responsibility, and melamine-laced raw materials sold as protein also reached formula makers directly (*The Adulteration of Milk With Melamine in China* F-3, R-3; Section 8.3).

## 4.3 Release: The Decision That Fixes the Exposed Population

The release of finished product is the plant's most consequential decision on the safety axis, because it fixes the exposed population. Most plants release on the strength of process controls validated to make food safe, supported by testing that samples rather than examines every unit. The class requires that a first- or second-band plant release only product whose process controls are shown to have operated within their limits for that product, on records the plant can trust, and, for a first-band product, only after testing whose design is stated against the hazards the plant's environment can carry. In the Sturgis case five environmental subsample surface swabs from the plant tested positive for *Cronobacter sakazakii*, and no finished product did (testimony of 26 May 2022); the hazard was in the plant, and the product sampling did not show it. A plant's environment is part of what its release decision rests on. A report from inside the plant reaches the release decision. A first- or second-band plant holds a channel by which any current or former worker can report a safety condition to the food-safety officer outside production management; a report is treated as a finding until it is resolved; and the time to detect of Section

3.3 is counted from it (*The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* F-1, R-1; noted to OpsCo).

## 4.4 Hold and Recall

When a hazard is found, the plant's ability to hold everything that may contain it, and to reach what it has already shipped, sets how much of the exposed population is protected. That ability rests on the plant's knowledge of which lots each input and each line entered, and on the distribution class's ability to find and withdraw them. The class requires that a first- or second-band plant be able to identify, within hours of a finding, every lot that may be affected and where it was shipped, and to hold every such lot still in its control; and that its records for doing so survive the loss of its information systems, because a hazard may be found while they are down (Section 5.5). The recall itself is carried out with the distribution class and the coordination class, and is theirs to state; the plant's part is knowing what to recall.

## 4.5 Substitution of Capacity

When a plant stops, the population is served from stock, from other plants and, for some commodities, from imports, and the plant's design basis states which. The class requires that a first-band plant's design basis state what would supply its dependent population if it stopped, how long that supply would last, and what would have to be true for it to be used, including the regulatory permissions that substitution requires. In 2022 the federal government flew infant formula into the country from abroad and relaxed import requirements to make up for the Sturgis stop, as reported. A substitution that depends on emergency action taken after the stop is a substitution the plant's design basis should state in advance, with the authority who would take it. The permissions substitution needs are in hand before the stop: for a first-band product the admission of substitute products, and its conditions, are granted or pre-authorized before any stop, with the authority who would invoke them named, because in 2022 the import flights and the enforcement discretion came about three months after the stop (*The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* F-2, R-2; noted to the framework for Section 10.6). It is the same finding *Ukraine's Payment System Through the War* makes of the permissions financial continuity needs.

## 4.6 Restart After a Safety Stop

A plant stopped for safety cannot restart until the hazard is found and corrected and the regulator is satisfied, and the time this takes is set by the investigation as much as by the repair. The class requires that a first-band plant hold, before it is ever stopped, the means to investigate its environment quickly, a plan for the corrections a hazard of each kind would require, and an agreed path with its regulator for resuming release, so that the time to restart of Section 3.3 is a planned figure and not a negotiated one. A first-band plant whose restart depends on a single site's facilities and utilities carries, in the Sturgis case's second stop, the reminder that a plant restarting under pressure is exposed to every other event of Section 5 at the same time.

# 5. Design Basis Events: The Envelope

The events below are the class's design-basis envelope, each stated with the event postulated, the consequence threshold, the function the class requires and the duration for which it is required. Sections 5.1, 5.2, 5.4 and 5.8 are non-adversarial; Section 5.3 may be either; Sections 5.5 and 5.7 are adversarial, and Section 5.6 is adversarial in its governing case but covers an adulterant that passes acceptance by accident as well; the adversarial events stand here because deliberate contamination at the processing chokepoint is among the framework's candidate bounding events in every domain, not only under military action (framework §4.1). Inclusion is by consequence severity, not likelihood.

## 5.1 Loss of the Plant by Fire or Natural Hazard

**Postulated event** fire, flood, wind or earthquake that destroys or disables a plant's process, refrigeration or utilities for longer than the commodity's supply-continuity duration.

**Consequence threshold** loss of the plant's share of national processing for its commodity until it is restored, and loss to its suppliers of product they cannot sell (Chain U).

**Required function** the plant states its time to restore against the supply-continuity duration, and what other capacity would take its supply meanwhile (Section 4.5); its fire protection and its siting are designed against the hazard; and its suppliers' fallback is stated.

**Design-basis duration** until the plant is restored to its design rate.

The fire at the Tyson Fresh Meats plant at Holcomb, Kansas, on 9 August 2019 is noted rather than proposed: it removed about six percent of national fed-cattle slaughter capacity for about four months, as reported, and wholesale beef prices rose while the price paid for cattle fell, the burden of a lost chokepoint falling on producers and consumers at once.

## 5.2 Loss of Power, Water or Refrigeration

**Postulated event** loss of grid power, municipal water or the plant's refrigeration for longer than its on-site provision, so that it cannot process, cannot clean or cannot keep product and raw material cold.

**Consequence threshold** loss of output (Chain U), and loss of product that warms beyond its limits, which must then be destroyed or, if it is not, becomes a safety hazard (Chain S).

**Required function** the plant states which product and raw material it can protect without each utility and for how long; it holds the means to stop receiving and to divert or destroy product safely; and it does not release product whose temperature history it cannot show.

**Design-basis duration** the longest outage the design-basis event can produce, as the energy and water series state it for the plant's location.

## 5.3 Release from the Ammonia Refrigeration System

**Postulated event** loss of containment from a plant's anhydrous-ammonia refrigeration system, through equipment failure, hydraulic shock, a fire or an attack on its controls.

**Consequence threshold** toxic exposure of the plant's workforce and of the public downwind, and loss of the plant's refrigeration (Chain U); the release consequence as the chemical series' shared model states it for the quantity and siting.

**Required function** the system meets the federal process-safety and risk-management requirements that apply to it and is designed against the release scenarios the shared model postulates; its relief, isolation and emergency shutdown functions act without depending on the plant's networked control system (Section 7.2); and the plant's emergency response is coordinated with the community's.

**Design-basis duration** until the release is stopped and the area is safe.

On the author's decision of 14 October 2026 the system is governed here and its release consequence is read through *Chemical Production and Synthesis* §4, cited and not restated. The ammonia release at the Millard Refrigerated Services warehouse in Theodore, Alabama, in August 2010, investigated by the Chemical Safety Board, in which more than 150 people working downwind were exposed, is noted as the kind of event the boundary covers; that facility was a refrigerated warehouse, which is the distribution class's, and the same boundary applies to it.

## 5.4 Pathogen in the Plant Environment and the Safety Stop

**Postulated event** a pathogen establishes itself in a plant's environment or process, reaches product, and is found through illness in the population, a regulator's finding or the plant's own testing, leading to a recall and a stop.

**Consequence threshold** illness and death in the exposed population (Chain S), and the loss of the plant's output for the time to restart (Chain U); for a first-band plant, the loss of supply to a population with no substitute.

**Required function** environmental monitoring designed against the pathogens the plant's product and process can carry, with its findings part of the release decision (Section 4.3); hold and recall within hours (Section 4.4); a stated substitution for a first-band product (Section 4.5); and a planned restart (Section 4.6).

**Design-basis duration** from introduction until the last affected lot is held or recalled; on the supply axis, until production resumes at the design rate.

The case is the shutdown of Abbott Nutrition's plant at Sturgis, Michigan, conferred by the framework as DBA-FA-RE-2 (framework §9.1), whose instrument reads it as non-adversarial. On the Food and Drug Administration Commissioner's account, the agency learned of a first *Cronobacter* infection in an infant who had consumed formula from the plant in September

2021; the Department of Health and Human Services' Office of Inspector General records two whistleblower complaints, one received in February 2021 that took the agency more than fifteen months to address and one received in October 2021 that was not escalated to senior leadership for nearly four months (report A-01-22-01502, 10 June 2024); the inspection began on 31 January 2022 after a delay caused by an outbreak of COVID-19 at the plant; the company stopped production about two days before the recall it began on 17 February 2022; five environmental swabs were positive and no finished product was; four infants were reported ill and *Cronobacter* may have contributed to two deaths; and a consent decree was entered on 16 May 2022 (testimony of 26 May 2022, Appendix A). The Commissioner stated that the timeline showed areas where the agency could and must do better or be faster. The design-basis reading is that a single plant's environment held a hazard its product sampling did not show, that the time to detect and hold was set by systems outside the plant as well as within it, and that the stop of a first-band plant was a national supply event that took months to restore. The record has since closed without a finding on causation. The Centers for Disease Control and Prevention closed their investigation, sequencing having found the patient samples not closely related to the plant's strains; the Department of Justice closed its criminal investigation, as reported in June 2026; and on 14 September 2026 the company settled the government's civil claims under the False Claims Act for about $385 million, the Department stating that the claims are allegations only and that there has been no determination of liability. Under the RECORD PRIMACY convention of the DBA Reference Event Register the class follows that record and makes no finding of fault or causation.

## 5.5 Cyber Attack on the Plant's Systems

**Postulated event** encryption, destruction or alteration of a plant's or its company's business and control systems, so that it cannot schedule, process, trace, label or release product.

**Consequence threshold** the stop of every plant that depends on the systems attacked (Chain U); and, where control systems or records are altered rather than encrypted, product released on false records (Chain S).

**Required function** the plant can stop safely and hold what it has; it can process, label and trace product for a stated period without its business systems, by means exercised in advance; its process controls and protective functions do not depend on the business network (Section 7.2); and it does not release product whose process records it cannot trust. The substitution is stated for the combined share of every plant that shares the systems attacked (Section 3.3), and restoration is credited only from backups and systems shown to be beyond the business network's reach, with the time to restore from them stated and exercised (*The Ransomware Attack on JBS* R-1, R-2).

**Design-basis duration** until the systems are restored and verified, or the plant's manual mode is relieved.

The case is the ransomware attack on JBS, the world's largest meat processor, conferred by the framework as DBA-FA-RE-1 (framework §9.1). The company found on 30 May 2021 that servers supporting its information systems in North America and Australia were under attack; its beef plants in the United States, and plants in Canada and Australia, stopped; the Federal Bureau of Investigation attributed the attack to the criminal group known as REvil; the company reported that production had resumed at all of its facilities within days and that less than one day's production was lost; and it later stated that it had paid a ransom of $11 million. The company's chief executive said the attackers had never reached its core systems, which greatly reduced the impact. The Department of Agriculture encouraged other processors to add capacity where they could. The attribution is stated as the Bureau's, and this instrument makes no finding of attribution. The design-basis reading is that an attack on the business systems of one company stopped its beef plants across a nation within a day, that the short duration rested on the attack not reaching the systems that mattered most, and that a plant's ability to work without its business systems is a design-basis requirement, not a convenience. The return in days rested on backups and core systems the attack did not reach, on the company's own statements, and its instrument finds that restoration is credited only from what an attack cannot reach (*The Ransomware Attack on JBS* F-2).

## 5.6 Adulteration of What the Plant Receives

**Postulated event** an adulterant or contaminant is added to raw material or an ingredient upstream of the plant, by a supplier, a hauler or a collection point, chosen or happening to pass the plant's acceptance tests.

**Consequence threshold** harm to the population that eats what is made from the pool (Chain S), at a scale set by the plant's throughput and distribution and by the number of plants that received from the same source; and the stop and recall that follow (Chain U).

**Required function** acceptance tests and supplier controls stated against the adulterants that would defeat them (Section 4.2); traceability from each lot of raw material to every lot it entered; and hold and recall within hours (Section 4.4). Acceptance tests the property and not only its proxy (Section 4.2); collection, hauling and aggregation are inside the plant's supplier controls (Section 4.2); and, with the coordination class, illness reports and surveillance of the dependent population are reviewed against the product promptly (*The Adulteration of Milk With Melamine in China* R-2).

**Design-basis duration** from the first adulterated delivery until every affected lot is held or recalled and the source is closed.

The case is the adulteration of milk with melamine in China in 2008, conferred by the framework as DBA-FA-RE-3 (framework §9.1). On the account of the Chinese authorities, as reported by the World Health Organization, melamine, an industrial chemical rich in nitrogen, was added to raw milk to make diluted milk appear to contain more protein; it reached infant formula and other dairy products of many companies, among them the Sanlu Group; and by the end of

November 2008, on the Ministry of Health's figures as the Chinese Center for Disease Control and Prevention's papers of 2009 give them, about 294,000 children had been diagnosed with urinary-tract stones, more than 50,000 had been hospitalized and six deaths had been confirmed. Melamine-laced raw materials sold as protein also reached formula makers directly, as the second of those papers adds. The adulteration was deliberate and economic rather than intended to harm, and the Chinese courts convicted those responsible. The design-basis reading is that an adulterant chosen to pass the processors' acceptance tests entered the pool upstream of them, that it reached a dependent population through a first-band product, and that the harm was national and delayed, as the framework's safety axis states (framework §1.5). That it was intended to profit rather than to harm does not change what reached the infants.

## 5.7 Intentional Adulteration Within the Plant

**Postulated event** a person with access to the plant, an insider or an intruder, introduces a contaminant intended to cause wide-scale harm at a point in the process after which it will not be removed.

**Consequence threshold** harm at population scale (Chain S), the delayed and attribution-resistant consequence the framework names (framework §1.5, §7); and the stop of the plant and its commodity's confidence (Chain U).

**Required function** the plant's vulnerability assessment, mitigation strategies and monitoring for the actionable process steps the federal intentional-adulteration rule requires (Section 8.4); access to those steps limited and recorded; and detection by the plant's process controls of a change at those steps that its operators would not otherwise see.

**Design-basis duration** from introduction until every affected lot is held or recalled and the population is warned.

No reference event is proposed for this category. The federal rule on mitigation strategies against intentional adulteration exists because the scenario is credible and has not, in the United States, been realized at scale; the class records it as a design-basis event without an empirical case of its own and reads it, under military action, at Section 6.3.

## 5.8 Loss of the Workforce

**Postulated event** illness, evacuation or the inability to reach the plant removes enough of the workforce that the plant cannot run.

**Consequence threshold** loss of the plant's output for the duration (Chain U).

**Required function** the plant states the staffing below which it cannot run safely and how it would reduce output in an orderly way above that, so that it stops without releasing product made outside its controls; and it holds the means to protect its workforce against the hazards the event brings.

**Design-basis duration** the duration of the event that removes the workforce.

The closures of meat and poultry plants in the spring of 2020, when the Centers for Disease Control and Prevention reported more than sixteen thousand cases of COVID-19 and 86 deaths among workers in 239 plants in 23 states by the end of May, and the Executive Order of 28 April 2020 delegating Defense Production Act authority to the Secretary of Agriculture to ensure that the plants continued to operate, are noted rather than proposed.

Table 5-1. Facility-class design-basis event categories.

| Event category | Postulated event (class-specific) | Consequence chain | Framework source |
|---|---|---|---|
| Loss of the plant by fire or natural hazard | Process, refrigeration or utilities destroyed beyond the supply-continuity duration | Chain U | §4.1 candidate (loss of concentrated processing capability) |
| Loss of power, water or refrigeration | Utility lost beyond on-site provision | Chain U + Chain S | §4.1 candidate (compound loss of a dependency); §6.1 |
| Release from the ammonia refrigeration system | Loss of containment of anhydrous ammonia | Release + Chain U | §6.2 (Chemical boundary); developed here |
| Pathogen in the plant environment and the safety stop | Pathogen reaches product; found through illness or inspection; recall and stop | Chain S + Chain U | developed here |
| Cyber attack on the plant’s systems | Business or control systems encrypted, destroyed or altered | Chain U (Chain S if altered) | developed here; §5 |
| Adulteration of what the plant receives | Adulterant added upstream, passing acceptance tests | Chain S + Chain U | §4.1 candidate (deliberate contamination at a chokepoint) |
| Intentional adulteration within the plant | Contaminant introduced at a step after which it is not removed | Chain S | §4.1 candidate; §7 |
| Loss of the workforce | Workforce unable to run the plant | Chain U | developed here |

# 6. Military-Action Extension: DBA-MA for the Processing Plant

## 6.1 MA Extension Applicability Determination

Every first- and second-band plant in the class performs the facility-level applicability determination the series makes mandatory, under the two-factor test of geolocation and strategic value (framework §7). The framework states a strong sector-level presumption toward applicability for concentrated processing plants (framework §7). For this class strategic value is driven by three things: the plant's supply share for its commodity; the dependence of a vulnerable population on what it makes; and the population its distribution reaches, which sets the scale of a contamination released from it. The determination is recorded, it is time-varying and re-triggered on the indicators the DBA-MA parent framework defines, and any scope reduction carries a recorded basis (Section 6.9).

## 6.2 The Four Characteristics Applied to the Class

Applied to the processing plant, the four military-action characteristics take specific form, in the canonical statement of FD-DBL (*Design Basis Layers*), at its §4.4. Intentionality: the adversary chooses the plant because its output feeds a population, a military force or a dependent group, or because a contamination released from it would reach a population the adversary wishes to harm or frighten. Persistence: the adversary holds access to a supplier, a plant's systems or a person inside it across the time needed to learn where in the process a contaminant would survive and when it would not be seen. Precision: the adversary chooses the commodity, the plant, the step and the day, so that the contaminant enters the pool after the kill step and before release, or so that the stop falls when stocks are lowest. Denial: the adversary takes the power, water, communications or laboratory capacity on which detection, hold and recall depend, so that a contamination reaches the population before it can be found, or strikes the plant itself.

Under the author's rulings of 10 and 13 October 2026, the military-action tier is not met without denial, and all four characteristics must be established on the record; a characteristic not determined counts as not present (*Design Basis Layers* §6.4). None of the three conferred events meets the military-action tier, on the determinations their instruments make at their §5.2: JBS and the melamine adulteration do not meet the threshold, each attributed to no state, and Sturgis is non-adversarial.

## 6.3 Deliberate Contamination at the Chokepoint, Timed to a Denial of Detection, as the Governing MA Scenario

The governing military-action scenario is a deliberate contamination introduced at a second- or first-band plant, at a step after which it will not be removed, by an adversary who at the same time denies the laboratory, communications and power on which the plant, the regulator and the public-health system depend to detect it, hold the affected lots and warn the population. It opens both chains. The contaminant is released and distributed before it is found; the plant's own

testing, the regulator's laboratories and the surveillance that would see illness in the population are slowed by the same campaign; and when it is found, the hold and recall must be run without the systems they normally use. No credit is taken for detection, traceability or recall that the same campaign can deny.

The extension requires three things. It requires that the plant's protection of its actionable process steps, the access controls and monitoring the intentional-adulteration rule requires, be designed against an insider or a supplier acting for an adversary, not only against a disgruntled individual. It requires that the plant's records of what entered each lot and where each lot went survive the loss of its information systems, its power and its communications, so that hold and recall can begin from the plant's own records (Section 4.4). And it requires that the plant be able to stop releasing on the first credible indication, before confirmation, under an authority named in advance, because a delay that waits on denied laboratories lets the exposed population grow.

The class has no empirical case of this scenario and proposes none. The melamine adulteration of 2008 shows the path, an adulterant chosen to pass acceptance tests entering the pool upstream and reaching a dependent population at national scale, without the intent to harm or the denial that would make it military action (Section 5.6). DBA-FA-UC-1 drives this scenario through the class (Section 9.2).

## 6.4 The Plant Struck

The second scenario is a strike on the plant itself, or on its power, water or refrigeration, to remove its share of a nation's food supply. A large processing plant is fixed, identifiable and, for its commodity, one of few, and an adversary seeking to reduce a population's food or a military force's supply can reach the processing chokepoint more easily than the many farms that feed it. The class requires a plant with a positive applicability determination to state what would supply its population if it were lost, which is the substitution of Section 4.5 read for a campaign rather than a single event, and to state the shelter and evacuation of its workforce, for whom the ammonia refrigeration system is itself a hazard when the plant is struck (Section 5.3).

## 6.5 The Coordinated Cyber Scenario

The governing cyber scenario is the JBS attack across more than one company at once, or against a company's control systems rather than its business systems. A single attack on one company's business systems stopped its beef plants across a nation within a day (Section 5.5). An adversary seeking denial rather than ransom would choose the companies that together carry most of a commodity, would reach the systems that control the process and its records, and would time the attack to a moment when stocks are low. Against it the class relies on the manual mode of Section 5.5, held ready and exercised; on the separation of the process controls and protective functions from the business network (Section 7.2); and on the refusal to release product whose process records cannot be trusted, which is the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*) applied to the release decision.

## 6.6 Cross-Sector Interaction

Under military action the processing plant is tied to other sectors in both directions. Coming in, an attacker may take the grid, which takes the plant's power and refrigeration; the water system, which takes its sanitation; the transportation system, which takes the arrival of animals and raw material and the departure of product; the chemical supply, which takes the sanitation chemicals and the ammonia its refrigeration needs; and the communications on which traceability and recall depend (framework §6.1). Going out, the plant's stop reaches its suppliers upstream at once and the population downstream within days, and a contamination it releases reaches the hospitals that must treat those it harms. The adversary need not attack the plant to stop it; an attack on what it consumes, at a time when stocks are low, is enough.

## 6.7 The Adequacy Result: Safe Supply Continuity Condition

The test of whether a plant's protection is adequate is whether it can reach what this document calls the Safe Supply Continuity Condition. When grid power, municipal water, the public networks, its own information systems and outside laboratory support are lost, the plant either continues to make food safely at a stated rate, or stops in an orderly way that releases nothing made outside its controls, holds everything that may be affected, and preserves the records from which every affected lot can be traced. Its kill steps and the protective functions of its refrigeration act on their own; its release decision rests on records it can trust; and its population is served, while it is stopped, from the substitution its design basis states in advance.

The protective functions on which this condition rests are to be locally sovereign as FD-LS defines the property where they are protective functions in that sense (*Local Sovereignty* §3.2): the flow-diversion device of a pasteurizer, which diverts product that has not reached its legal temperature and time; the temperature and time controls of a retort; and the pressure relief, isolation and emergency shutdown of the ammonia refrigeration system. Each is to be shown to be a measurement against a setpoint, colocated with what it protects, and no system an attacker could reach should be able to stop, delay or trigger it, or alter its settings except through a governed act. Local Sovereignty governs and is not restated here. It describes what protects the plant's process and its people when the plant's control systems have failed; it does not decide whether any action needs human authorization (*Local Sovereignty* §4).

The end state may also be reached by stopping. A plant that stops in an orderly way, holding what it has made, ends its safety exposure at the cost of its supply share. Like the drawdown of a dam or the evacuation of a hospital, it is a protective action with an authority named in advance, a trigger stated in advance, and a statement of what it costs and to whom, the population that loses its supply and the producers upstream who lose their market.

## 6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Safe Supply Continuity Condition is achievable with the Section 4 and Section 5 requirements: acceptance tests stated against what would defeat them,

release on trusted records, hold and recall within hours, a stated substitution for first-band products, a planned restart, and a manual mode for the loss of information systems. Against a deliberate adversary, adequacy has limits.

Three residuals are declared, in the output form DBA-MA-DiD (*Defense in Depth*) sets at its §3.5. First, a contaminant chosen to pass every acceptance test and introduced upstream of the plant reaches the pool; beyond the design act the residual is contaminated product released until the population's illness reveals it, and the response is the speed of detection and recall, not the prevention of release. Second, an insider at an actionable process step can defeat the access controls the rule requires; the residual is a deliberate contamination the plant's monitoring may not see, and the response is the plant's process controls detecting the change and the population-health system detecting the harm. Third, the loss of a first-band plant for longer than its substitution can bear leaves a dependent population without its food; beyond that point the residual is harm to that population, and the design response is a supply that does not rest on one plant, which no single plant's design basis can provide. These residuals bound the class adequacy claim.

### 6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as an exposed population credited on a time to detect the regulator rather than the plant controls, a substitution credited that depends on emergency action not yet taken, a manual mode credited that has not been exercised, an acceptance test credited against an adulterant it does not detect, or a time to restart credited without the regulator's agreement, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Sections 9.1 and 10.10 record it for the framework to adopt. A restart credited on the assumption that core systems survive an attack is such a relaxation and needs a basis (*The Ransomware Attack on JBS* R-2).

## 7. The Dual Design Basis, Stated in Full

### 7.1 The Two Design Bases

The plant carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4.

### 7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the plant relies on to make food safely and to protect its people independently of its networked systems. These are the kill steps and the controls that enforce them, pasteurization, retorting and cooking to validated times and

temperatures; the flow-diversion device of a high-temperature, short-time pasteurizer, which the Grade A Pasteurized Milk Ordinance requires to divert milk that has not reached its required temperature; the relief, isolation and emergency shutdown of the ammonia refrigeration system; and the integrity of the records on which the release decision rests. A flow-diversion device that measures the temperature of the product at the end of the holding tube and diverts it on its own measurement, with no networked controller able to inhibit or command it, is the class's clearest candidate for a locally sovereign function. A function meets *Local Sovereignty* §3.2 only if no system an attacker could reach can stop, delay or trigger it, or change its settings except through a governed act. A plant control system that can change a pasteurizer's setpoint from the plant's business network fails that test for every function it can change.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Controllers that behave predictably within a stated range of inputs, and that have been formally tested in the deployed configuration, are outside the Command Broker obligation for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line: a pasteurizer's controller takes, in deciding whether milk is pasteurized, an action whose consequence can sicken a population, and it is outside the obligation only because, and only while, its behavior is bounded by that showing (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act (Determination of 8 October 2026, A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out; *Chemical Production and Synthesis* §7.2). A recipe or setpoint pushed to a plant's controllers over the network is such a change. A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence.

## 7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are entering the processing plant: vision systems that inspect carcasses and product for defects and contamination, systems that grade and sort, systems that predict when a line or environment will fail its sanitation, systems that formulate products and adjust process parameters, and systems that analyze test results and recommend whether a lot should be released. The framework names automated processing control and contamination-screening dispositions among the actions to be placed (framework §5). These systems are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*); CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The coordination class, OpsCo, is the locus at which the sector's placement and obligation are implemented (framework §3.5). Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3), and the qualified operator or food-safety officer who authorizes an above-line action is the Command Broker the framework names (framework §5).

Four cases illustrate the application, and none of them is a placement by component. A system within scope that recommends the release of a lot, together with its acceptance, is one action placed on the consequence of the release, which is set by the exposed population of Section 3.3: a release whose consequence can reach the threshold is above the line and requires the authorization of the plant's food-safety officer, given on a basis the system cannot alter. A system within scope that changes a process parameter at a kill step takes an action placed on the consequence of the change, which can be as severe as an unpasteurized release; a change within limits the plant's process authority has validated may be discharged in advance under FD-BL-D1 (*Mode of Discharge*), where the validated limits form a pre-approved envelope and the controller that enforces them is verified, and a change to those limits is excluded from advance discharge. A vision system that removes a carcass or a product from the line takes an action placed on the consequence of the removal, and one that passes a carcass it should have removed takes the action of passing it. And a system that formulates a first-band product takes an action placed on the consequence to the population that depends on it.

Any output on which an above-line action is grounded carries the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*): a release recommendation computed from process records that may have been altered is not a basis for authorization until the records can be trusted (Section 6.5).

## 7.4 The Interaction Point: The Release Under Pressure

The two design bases meet when the plant must decide whether to release while its systems are under attack or its output is urgently needed. The plant's control systems must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That obligation reaches each such action whose consequence meets the threshold, and the actions most likely to do so are the release of a lot to a large or dependent population, a change to a kill step's parameters, and a change to the protective functions of the refrigeration system; each is placed on its own consequence. And the plant's protection must hold when no outside system can be reached at all. The physical design basis guarantees the kill step and the refrigeration's protection; the manual mode guarantees that the plant can stop and hold; the AI-governance design basis guarantees that no proposing system can release a lot or change a kill step whose consequence meets the threshold without a qualified human authorization the architecture can enforce, discharged in the moment or, where FD-BL-D1 (*Mode of Discharge*) permits, in advance. When the systems fail, the plant's food-safety officer decides by hand, and the default when records cannot be trusted is to hold.

## 7.5 Scope Boundary

This instrument states the physical design basis in full and names the AI-governance design basis without restating it. The Foundational Definitions own the Bright Line, its scope and its obligation; the Command Broker family owns the Command Broker; the coordination class owns

the implementation of placement in the sector; and the agricultural-production and distribution classes own the placement of actions in their facilities. This instrument applies them to the processing plant.

# 8. Cross-Sector Dependencies and Regulatory Context

## 8.1 What Other Sectors Depend On: the Processing Plant

Every sector's workforce and population eat what this class makes, and the framework calls food a universal dependency whose failure is felt across every other sector (framework §6.3). Three dependencies are particular. Hospitals and their patients depend on first-band products, infant formula and medical foods, that the healthcare series does not make and cannot substitute. Military forces and emergency services depend on the plants' output in a crisis. And the producers upstream, the farms and feedlots of the agricultural-production class, depend on the plants as their only market for perishable output, so that a plant's stop is felt first by them.

## 8.2 What the Plant Depends On

The plant depends on the five-sector web the framework names (framework §6.1): energy for its process and refrigeration; water for its sanitation and process; transportation for its inflow of animals and raw material and its outflow of product; the chemical sector for its sanitation chemicals and the ammonia its refrigeration needs; and financial services for the payments to its suppliers and from its customers. It depends also on the information-technology series for the software of its business and control systems, whose supply chain is *Software, Hardware and Product Supply Chain* (DBA-IT-SW-FC3), and on the communications series for the traceability and recall that depend on reaching its customers. The compound loss of more than one of these is the framework's governing inward case (framework §6.1), and the governing military-action scenario of Section 6.3 is one form of it.

## 8.3 The Boundaries with the Partner Classes and Series

Five boundaries are drawn. With *Agricultural Production* (DBA-FA-AG-FC1): the farm, the feedlot, the dairy herd and their health are that class's, and the animals and raw material are this class's from the moment the plant accepts them; highly pathogenic avian influenza, which the plan for closing the seven sectors named for this sector, is held for that class (Section 9.2). With *Storage, Distribution and Retail* (DBA-FA-DR-FC3): finished product is that class's from the plant's dock, and the recall is carried out with it. With the coordination class, OpsCo: the inspectors, the regulatory framework and disease surveillance are its members, and the plant's response to them is this class's. With the chemical series: the sanitation chemicals and the ammonia a plant buys are produced there, and the plant's ammonia refrigeration system is governed here with its release consequence read through *Chemical Production and Synthesis* §4, on the author's decision of 14 October 2026. With the healthcare series: a medical food or infant formula made in a processing plant is this class's product, and its consumption by patients is the

care-delivery class's concern. The boundary with *Agricultural Production* is drawn so that no stage between farm and plant falls between classes: collection stations, hauling and aggregation are within this class's supplier controls from the moment raw material is consigned to the plant, whatever class holds the farm (*The Adulteration of Milk With Melamine in China* R-3; noted to *Agricultural Production*).

## 8.4 Regulatory and Institutional Context

The class is governed under the split jurisdiction the framework describes (framework §2.1). Meat, poultry and processed-egg plants operate under continuous inspection by the Department of Agriculture's Food Safety and Inspection Service, under the Federal Meat Inspection Act, the Poultry Products Inspection Act and the Egg Products Inspection Act, and must hold hazard analysis and critical control point plans (9 CFR Part 417). Other processing plants are regulated by the Food and Drug Administration under the Food Safety Modernization Act of 2011, whose preventive-controls rule requires a written food-safety plan with a hazard analysis and preventive controls (21 CFR Part 117), and whose rule on mitigation strategies to protect food against intentional adulteration, finalized in May 2016, requires the larger facilities to assess their vulnerable process steps and put mitigation strategies in place (21 CFR Part 121). Infant formula carries its own requirements for current good manufacturing practice, quality control and records (21 CFR Parts 106 and 107). Pasteurized milk is produced under the Grade A Pasteurized Milk Ordinance, adopted by the states. A plant's ammonia refrigeration system above the threshold quantity falls under the Occupational Safety and Health Administration's process safety management standard (29 CFR 1910.119) and the Environmental Protection Agency's risk management program (40 CFR Part 68).

The regulatory gap is the framework's (framework §2.3). The regimes state process controls, plans and inspections, and the intentional-adulteration rule states a vulnerability assessment; none states the set of events a plant must be shown to withstand, expressed as a supply-continuity duration and an exposed population; none requires a first-band plant to state in advance what would supply its dependent population if it stopped, which the Commissioner's testimony of 2022 recommended in part by asking for notification of supply interruptions and for risk-management plans; and none treats the plant's ability to stop, hold and trace without its information systems as a requirement. This class states them as design-basis requirements for the bands that carry the load.

# 9. Inheritance Map and Anticipated Derivatives

## 9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments inherit the framework's methodology and architecture and the water sector's contamination-consequence structure on the safety axis, and add the food-function-specific

consequence chain on both axes (framework §8.1). Table 9-1 is the map for this class. Where the framework is a skeleton, the element is developed here and recorded for the framework to take up.

Table 9-1. Inheritance map for DBA-FA-FP-FC2.

| Element | Source in the framework | Inheritance mode |
|---|---|---|
| Function-primary placement, FP function class | §1.3, §3.1, §3.3 | Inherited unchanged |
| Commodity and perishability sub-index | §3.1, §8.2 | Inherited with adaptation; three bands stated, beverage and packaged food at sub-index depth (Section 2.2) |
| Design-basis principles (consequence severity, not likelihood) | §4.1 | Inherited unchanged |
| Dual consequence: supply-continuity duration and exposed population | §3.6 (proposed) | Calibrated here as five parameters, the two axes kept separate (Section 3.3); for the framework to take up |
| Contamination-consequence structure on the safety axis | §1.5, §8.1 (from the water series) | Inherited unchanged |
| Event categories | §4.1 (candidates) | Developed here ahead of the framework; for the framework to take up |
| Bright Line and Command Broker architecture | §5; FD-BL; Command Broker family | Inherited with adaptation: the release decision placed by exposed population (Section 7.3) |
| Release, hold and restart | §3.3 | Locally defined (Section 4) |
| Ammonia refrigeration and the chemical release model | §6.2 (Chemical boundary) | Locally governed; release consequence cited from the chemical series (Section 5.3) |
| Safe Supply Continuity Condition | §7 (skeleton) | Locally defined (Section 6.7) |
| Declared residuals and conditionality check | not yet in the framework | Locally defined; recorded for the framework (Sections 6.8, 6.9) |
| Five-sector dependency web | §6.1, §6.2 | Inherited and extended to information technology and communications (Section 8.2) |

## 9.2 Anticipated Derivatives

Plant-, company- and commodity-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. The three reference events this class proposed at v0.1, on the author's decision of 14 October 2026, are conferred by the framework at §9.1: DBA-FA-RE-1, *The Ransomware Attack on JBS*, May and June 2021 (Section 5.5); DBA-FA-RE-2, *The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage*, 2022 (Section 5.4); and DBA-FA-RE-3, *The Adulteration of Milk With Melamine in China*, 2008 (Section 5.6). Their instruments were written by Project 11 on 14 October 2026, and this revision takes in their erratum and seven recommendations. Highly pathogenic avian influenza, which the plan for closing the seven sectors named for this sector, is held for *Agricultural Production* on the author's decision of the same date. The Holcomb plant fire of 2019, the Millard ammonia release of 2010 and the closures of 2020 are noted in Section 5 and not proposed. The facility-class use case for the governing military-action scenario of Section 6.3 is DBA-FA-UC-1, issued under this revision in the fifth wave of the plan.

# 10. Gaps and Open Items

## 10.1 Quantitative Design Parameters

The five parameters of Section 3.3 are stated without values. Their values for a plant are to be validated against its production and shipment records, its food-safety plan and hazard analysis, its testing and environmental-monitoring records, its recall history and the regulator's findings, and, for the supply share and continuity duration, against the national data the Department of Agriculture publishes for each commodity.

## 10.2 The Dual Consequence Measure

The framework proposes a supply-continuity duration and a contamination consequence measured by exposed population, and asks that they not be collapsed (framework §3.6). Section 3.3 calibrates them as five parameters, supply share, supply-continuity duration, exposed population, time to detect and hold, and time to restart, and keeps the axes separate. Whether the framework adopts the five for the sector is the framework's to decide, and the agricultural-production and distribution classes should be read against them before it does.

## 10.3 Beverage and Packaged-Food Manufacture

On the author's decision of 14 October 2026, beverage and packaged-food manufacture are stated at the depth of the sub-index. A later revision is to develop them, with particular attention to the packaged-food plant that combines ingredients from many suppliers, whose acceptance problem is wider than the dairy's (Section 4.2), and to the low-moisture products in which pathogens survive.

## 10.4 The Ammonia Refrigeration Boundary

On the author's decision of 14 October 2026 the plant's ammonia refrigeration system is governed here and its release consequence is read through *Chemical Production and Synthesis* §4. That section holds the shared chemical-release model provisionally, and if it moves the citation follows it. The chemical series should be told, when this class is conferred, that a processing plant's refrigeration system now cites its model, and the refrigerated warehouses of the distribution class will cite it the same way.

## 10.5 Local Sovereignty in the DBA-MA Sense

Section 6.7 relies on locally sovereign protective functions, the pasteurizer's flow diversion, the retort's controls and the refrigeration's relief and shutdown, as FD-LS defines the property. Whether the DBA-MA parent framework's local-sovereignty principle is the same thing FD-LS defines remains the open question the other classes record.

## 10.6 First-Band Substitution

Section 4.5 requires a first-band plant to state what would supply its dependent population if it stopped. No single plant can provide the substitution, which rests on other makers, on stocks and on the regulator's permissions; the requirement states what the plant must know, not what it must hold. Whether the coordination class or the framework should state a sector requirement for the substitution of first-band products is for the framework. Section 4.5 now requires also that the permissions substitution needs be in hand before the stop, which *The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* R-2 notes to the framework for this question.

## 10.7 The Reference Events

The three reference events this class proposed are conferred by the framework at §9.1 as DBA-FA-RE-1 to DBA-FA-RE-3, and their instruments were written by Project 11 on 14 October 2026. DBA-FA-RE-1 records JBS as not meeting the threshold, the Bureau's attribution to a criminal group stated as the Bureau's; DBA-FA-RE-2 records Sturgis as non-adversarial, with Healthcare and Public Health an impacted sector and the plant's causal role unresolved on a record that has since closed (Section 5.4); and DBA-FA-RE-3 records the melamine adulteration as not meeting the threshold, a crime prosecuted in China and attributed to no state, with Healthcare and Public Health an impacted sector. The instruments name institutions and not individuals, and the melamine instrument reads the Chinese authorities' own figures. A public body's figure for the share of JBS in beef capacity was not found, and DBA-FA-RE-1 holds that open. This revision takes in the erratum at Section 5.4 and the recommendations at Sections 3.3, 4.2, 4.3, 4.5, 5.5, 5.6, 6.9 and 8.3. The use case for the governing scenario of Section 6.3 is DBA-FA-UC-1.

## 10.8 The Register Row

The DBA Facility Class Register now records this class at v0.1 as an issued working draft under the framework at v0.4; Project 0 entered the row on 14 October 2026.

## 10.9 The Order of the Sector's Classes

This class is numbered FC2 by the framework and is written first, because it carries the sector's chokepoint. The framework's numbering records conferral, not order of writing, and no renumbering is proposed.

## 10.10 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the five parameters of the dual consequence (Section 3.3, Section 10.2), the event categories of Section 5, the three bands of the sub-index (Section 2.2), the declared residuals of Section 6.8, the conditionality principle of Section 6.9, and the substitution question of Section 10.6, with *The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* R-2 beside it; the conferral section it also asked for is now framework §9.1. To *Agricultural Production*, when written: the event held for it and the boundary of Section 8.3, drawn so that no stage between farm and plant falls between classes (*The Adulteration of Milk With Melamine in China* R-3). To OpsCo, when written: the inside report reaching the release decision (*The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* R-1) and the watch on the dependent population's health (*The Adulteration of Milk With Melamine in China* R-2). To the chemical series: that this class cites its shared release model for ammonia refrigeration (Section 10.4). To the healthcare series: that first-band products, infant formula and medical foods, are made in this class and their loss is a supply event for patients (Section 8.1), for *Care Delivery* to read, with the watch on the dependent population's health (*The Adulteration of Milk With Melamine in China* R-2), which reaches clinicians. To the water series: that a processing plant's sanitation depends on municipal water and its loss stops the plant (Section 5.2). The register row is entered (Section 10.8).

# References

## DBA-FA Series and Parent Framework

*Food and Agriculture Sector Framework* (DBA-FA_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the design-basis-under-military-action methodology. FD-DBL (*Design Basis Layers*), held for the edition following the October 2026 Edition, is the canonical statement of the four military-action characteristics (§4.4, §6.4).

*Defense in Depth* (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

*Chemical Production and Synthesis* (DBA-CHEM-PR-FC1), *Care Delivery* (DBA-HPH-CD-FC1) and *Software, Hardware and Product Supply Chain* (DBA-IT-SW-FC3). Eclectic Technologies, 2026. The structural model for this instrument, the shared chemical-release model it cites, and the supply chain of its systems.

*The Ransomware Attack on JBS* (DBA-FA-RE-1), *The Shutdown of Abbott Nutrition's Sturgis Plant and the Infant-Formula Shortage* (DBA-FA-RE-2) and *The Adulteration of Milk With Melamine in China* (DBA-FA-RE-3). Eclectic Technologies, 2026. Held by Project 11; conferred at Food and Agriculture Sector Framework §9.1 and cited by short title and finding or recommendation number. *Ukraine's Payment System Through the War* (DBA-FIN-RE-3), cited for its parallel finding.

*Facility-Class Types Are Physical, Coordination and Logical*, author's determination of 30 September 2026, and *A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out*, author's determination of 8 October 2026, both recorded by Project 0. *Closing the Seven Sectors*, working plan of 13 October 2026, Project 9.

## Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

## Regulatory, Standards and Guidance

Federal Meat Inspection Act; Poultry Products Inspection Act; Egg Products Inspection Act; 9 CFR Part 417, Hazard Analysis and Critical Control Point Systems. FDA Food Safety Modernization Act, 2011; 21 CFR Part 117, Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food; 21 CFR Part 121, Mitigation Strategies to Protect Food Against Intentional Adulteration, final rule, May 2016; 21 CFR Parts 106 and 107, infant formula. Grade "A" Pasteurized Milk Ordinance, Food and Drug Administration and the states.

Occupational Safety and Health Administration. Process Safety Management of Highly Hazardous Chemicals, 29 CFR 1910.119. Environmental Protection Agency. Risk Management Program, 40 CFR Part 68.

## Events and Investigations

Food and Drug Administration. Testimony of the Commissioner before the Senate Committee on Health, Education, Labor and Pensions on the infant formula supply, 26 May 2022, with Appendix A, timeline. Contemporary reporting on the restart and flooding of the Sturgis plant,

June 2022. Department of Health and Human Services, Office of Inspector General. *The Food and Drug Administration's Inspection and Recall Process Should Be Improved To Ensure the Safety of the Infant Formula Supply*, A-01-22-01502, 10 June 2024. Department of Justice, release of 14 September 2026 on the settlement of civil claims.

JBS USA. Statements of 31 May to 9 June 2021. Federal Bureau of Investigation. Statement on the attribution of the JBS attack, 2 June 2021. United States Department of Agriculture. Statements on the JBS attack, June 2021. Reporting by Feedstuffs and others, June 2021. Bloomberg, report of 2 June 2021.

World Health Organization. Reports on melamine in milk products in China, 2008. Contemporary reporting of the Chinese Ministry of Health's figures, December 2008. Chinese Center for Disease Control and Prevention, Institute of Nutrition and Food Safety, two papers in *Biomedical and Environmental Sciences*, 2009, as read in DBA-FA-RE-3.

American Farm Bureau Federation. *Impacts of the Packing Plant Fire in Kansas*, Market Intel, 2019. United States Chemical Safety and Hazard Investigation Board. Investigation of the ammonia release at Millard Refrigerated Services, Theodore, Alabama, August 2010. Centers for Disease Control and Prevention. *Update: COVID-19 Among Workers in Meat and Poultry Processing Facilities — United States, April–May 2020*, Morbidity and Mortality Weekly Report, July 2020. Executive Order 13917, *Delegating Authority Under the Defense Production Act With Respect to Food Supply Chain Resources During the National Emergency Caused by the Outbreak of COVID-19*, 28 April 2020.

# Revision History

| Version | Date | Author | Description |
|---|---|---|---|
| 0.1 — DRAFT | October 2026 | Tim Roxey | Initial issue. First facility-class instrument in the DBA-FA series, typed physical under the author's determination of 30 September 2026, in the ten-section structure of DBA-HPH-CD-FC1 and DBA-CHEM-PR-FC1. Written under the plan Closing the Seven Sectors, wave 2, on the author's decisions of 14 October 2026: all processing commodities at class level with three reference facilities, beverage and packaged food at sub-index depth; the plant's ammonia refrigeration governed here with its release consequence read through the chemical series' shared model; three reference events proposed, JBS (2021), Abbott Sturgis (2022) and the melamine adulteration of milk (2008), with highly pathogenic avian influenza held for Agricultural Production. Calibrates the framework's dual consequence as five parameters with the two axes kept separate, states release, hold and restart as the governing pathway, and the Safe Supply Continuity Condition. Checked before issue by an independent review agent, not a person, against primary sources and for internal consistency; its findings were taken in, among them the Sturgis counts restated as the |

| Version | Date | Author | Description |
|---|---|---|---|
| | | | Commissioner gave them with the plant's causal role unresolved, and the interaction point restated by consequence. |
| 0.2 — DRAFT | October 2026 | Tim Roxey | Conforms to the framework at v0.5 and takes in the return of Project 11's Food and Agriculture batch. The three reference events proposed at v0.1 are stated as conferred, as DBA-FA-RE-1 (JBS), DBA-FA-RE-2 (Sturgis) and DBA-FA-RE-3 (melamine), at Sections 1.3, 5.4, 5.5, 5.6, 6.2, 9.2 and 10.7. Erratum 1 gives both whistleblower complaints the Inspector General records, with its report, and records the closing of the Sturgis record under RECORD PRIMACY (Section 5.4). The optional refinements give the JBS share of beef capacity and the fall in slaughter with their sources (Section 1.3) and the melamine figures from the Chinese authorities' own papers (Sections 2.4, 5.6). The seven recommendations enter at Sections 3.3 (the supply share across the shared system; time to detect from the inside report and the watch on the dependent population), 4.2 (the property, not the proxy; the stretch between farm and plant), 4.3 (the inside report reaches the release decision), 4.5 (substitution permissions before the stop), 5.5 and 6.9 (restoration credited only from what the attack cannot reach), 5.6 and 8.3. Section 10.10 carries the notings to OpsCo, Care Delivery, Agricultural Production and the framework. The cover lineage stays as issued. No change to the class's scope, chains, parameters, event categories or residuals; the v0.1 entry is left as a record. |